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Experiment No. 1: Comparative Study of Low-Code Development Versus Traditional Application Development Approaches

1. Aim

To study and compare the **Traditional Application Development** approach with the **Low-Code Development** approach based on their development process, speed, cost, flexibility, and ease of use.

2. Introduction to Traditional Application Development

Traditional application development is the conventional method of building software by writing source code using programming languages such as Java, Python, C#, or
JavaScript. Developers manually design, code, test, deploy, and maintain the application.

This approach provides complete control over the application's functionality, performance, and customization. However, it requires skilled developers, longer development time, and higher development costs.

Features

- Manual coding is required.
- High flexibility and customization.
- Suitable for complex applications.
- Requires experienced developers.
- Longer testing and deployment cycles.

3. Introduction to Low-Code Development

Low-code development is a modern software development approach that enables applications to be built using graphical user interfaces, drag-and-drop components, and minimal hand-written code.

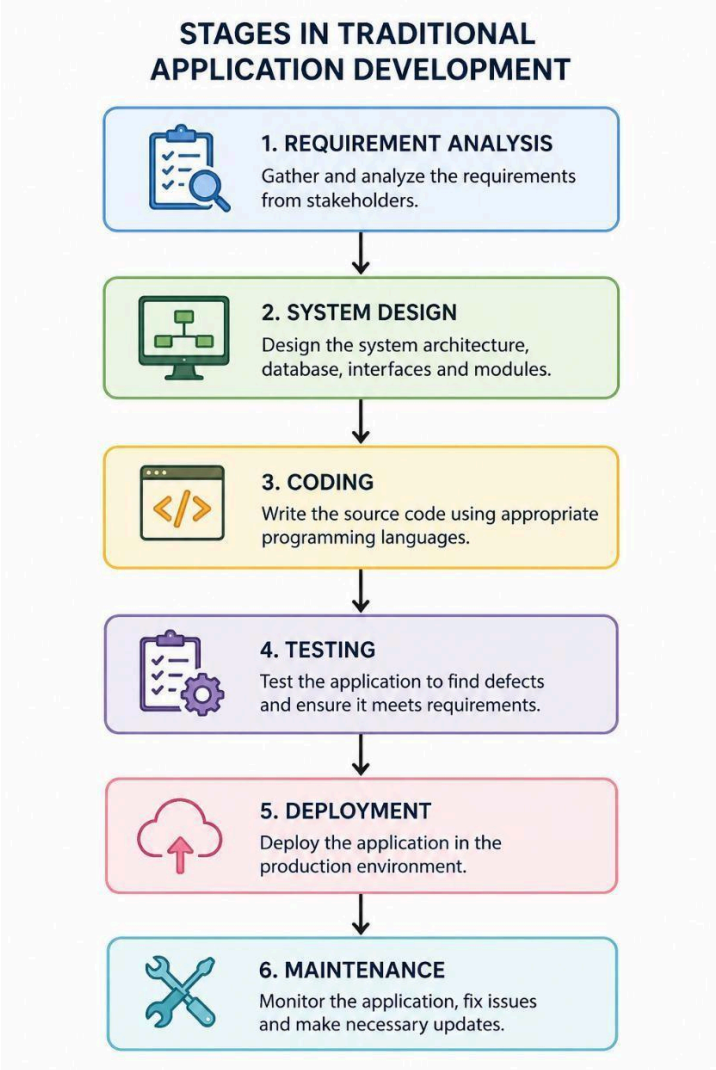
It allows both professional developers and non-technical users (citizen developers) to create applications quickly, reducing development time and effort.

Examples of low-code platforms include Microsoft Power Apps, Mendix, OutSystems, and Appian.

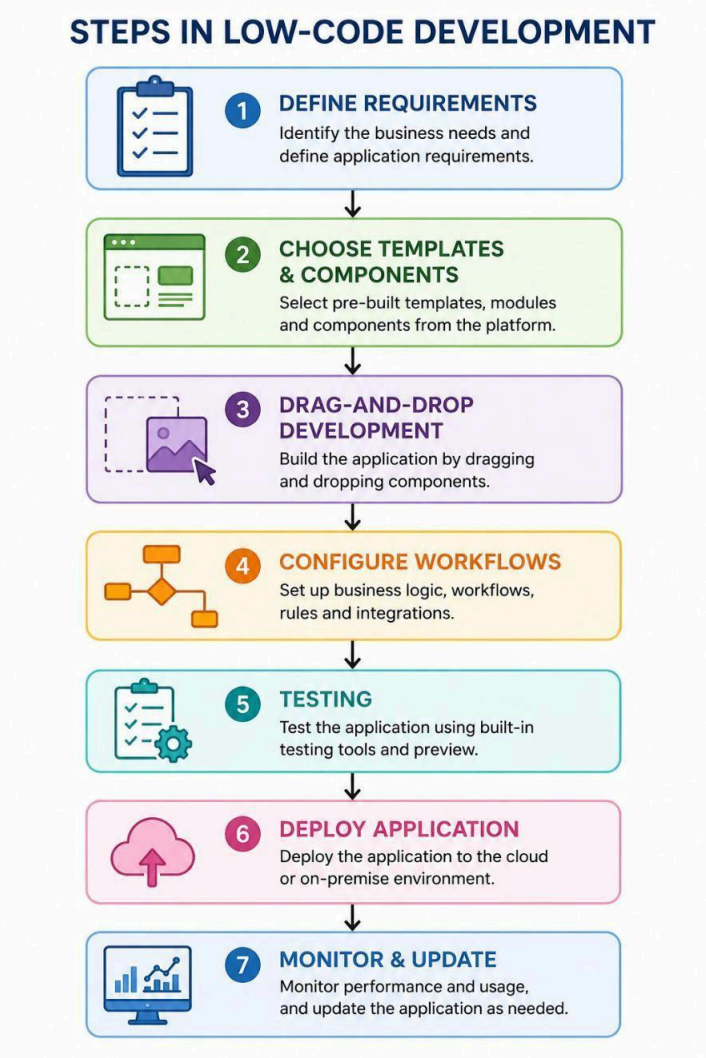
Features

- Drag-and-drop interface.
- Minimal coding required.
- Faster application development.
- Easy integration with existing systems.
- Suitable for rapid application development and business applications.

4. Diagram for Stages in Traditional Application Development



5. Diagram for Steps in Low-Code Development



6. Comparative Analysis of Traditional and Low-Code Development

Parameter	Traditional Development	Low-Code Development
Coding	Extensive manual coding	Minimal coding
Development Speed	Slow	Very Fast
Development Cost	High	Low
Required Skills	Professional programmers	Developers and non-developers

Customization	Very High	Moderate to High

Parameter	**Traditional Development**	**Low-Code Development**
Learning Curve	Steep	Easy
Maintenance	Time-consuming	Easier
Testing	Mostly manual	Automated tools available
Deployment	Takes more time	Quick deployment
Best Suitable For	Large and complex applications	Business applications and rapid development

7. Result

The comparative study shows that **Traditional Application Development** offers maximum flexibility and customization but requires more time, skilled developers, and higher costs.

On the other hand, **Low-Code Development** significantly reduces development time, simplifies application creation, and lowers development costs, making it ideal for rapid application development and business solutions. The choice between the two approaches depends on the project's complexity, budget, and development timeline.